Chapter Three
Theory of Consumer Behavior

Introduction

Consumer behavior can be best understood in three steps.

1. By examining consumer ‘s preference, we need a practical way to describe how people prefer one good to another
2. we must take into account that consumers face budget constraints they have limited incomes that restrict the quantities of goods they can buy.
3. we will put consumer preference and budget constraint together to determine consumer choice.

3.1 Consumer preferences

Consumer preference refers the relative attitudes of a consumer towards different consumption bundles.

If any two consumption bundles: **A and B** from a given consumption set, is given to a consumer, then the consumer can make either of the following statements.

- He may say he would prefer to have **A** than **B** without any ambiguity it is known as **strict preference.**

A>B , this implies that consumption bundle **A** is more preferable than **B** by the consumer.

- He may say he prefers to have **A** than **B** or is indifferent between having **A** and **B** it is known as **weak preference.**

A≥B, this implies that consumption bundle **A** is at least as good to the consumer as **B.**

- He may say he is indifferent between having **A** or **B it** is known as **indifference preference.**

A~ B, this implies that the consumer is indifferent between A and **B.**

3.2 The concept of utility

utility to describe the satisfaction or pleasure derived from the consumption of a good or service.

In defining utility, it is important to bear in mind the following points.

Utility' and 'Usefulness' are not synonymous. Attachment of high utility to a commodity by a consumer may not necessarily indicate the biological or social usefulness of the commodity.

· ***Utility is subjective***. The utility of a product will vary from person to person.

Utility can be different at different places and time.

3.3 Approaches of measuring utility

1. **Cardinal** : utility can be **measured objectively**.
2. **Ordinal approaches**. utility is not **measurable in cardinal numbers** rather the consumer can rank or order the utility he derives from different goods and services.

3.3.1 The cardinal utility theory

According to the cardinal utility theory, utility is measurable by arbitrary unit of measurement called **utils** in the form of 1, 2, 3 etc. For example, we may say that consumption of an orange gives Bilen 10 utils and a banana gives her 8 utils, and so on. From this, we can assert that Bilen gets more satisfaction from orange than from banana.

3.3.1.1 Assumptions of cardinal utility theory

The cardinal approach is based on the following major assumptions.

1. Rationality of consumers. The main objective of the consumer is to maximize his/her satisfaction given his/her limited budget or income.

2. Utility is cardinally measurable.

3. Constant marginal utility of money.

4. Diminishing marginal utility (DMU). The utility derived from each successive units of a commodity diminishes.

5. The total utility of a basket of goods depends on the quantities of the individual commodities.

total utility is given by TU = f (X_1 , X_2 .....X_n).

3.3.1.2 Total and marginal utility

Total Utility (TU) is the total satisfaction a consumer gets from consuming some specific quantities of a commodity at a particular time.

Marginal Utility (MU) is the extra satisfaction a consumer realizes from an additional unit of the product. In other words, marginal utility is the change in total utility that results from the consumption of one more unit of a product. Graphically, it is the slope of total utility.

Mathematically, marginal utility is:

$$MU = \frac{\Delta TU}{\Delta Q}$$

To explain the relationship between TU and MU, let us consider the following hypothetical example.

Table 3.1: Total and marginal utility

Quantity	Total utility (TU)	Marginal utility (MU)
0	0	-
1	10	10
2	18	8
3	24	6
4	28	4
5	30	2
6	30	0
7	28	-2

Graphically, the above data can be depicted as follows.

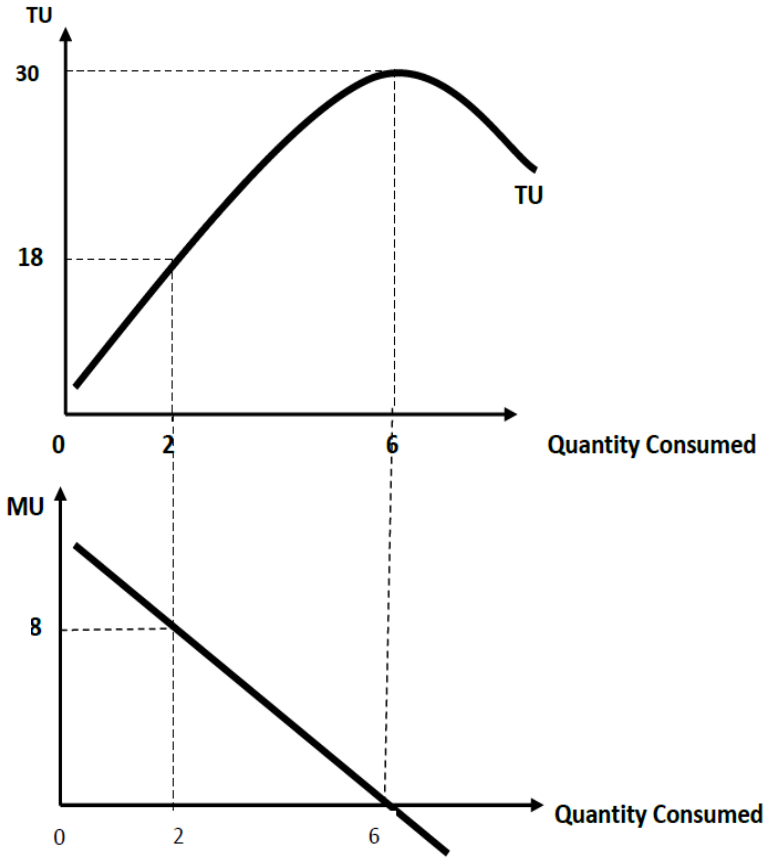


Figure 3.1: Total and marginal utility curves

As it can be observed from the above figure,

- When TU is increasing, MU is positive.
- When TU is maximized, MU is zero.
- When TU is decreasing, MU is negative.

3.3.1.3 Law of diminishing marginal utility (LDMU)

The *law of diminishing marginal utility* states that as the quantity consumed of a commodity increases per unit of time, the utility derived from each successive unit decreases, consumption of all other commodities remaining constant.

In other words, the extra satisfaction that a consumer derives declines as he/she consumes more and more of the product in a given period of time.

The law of diminishing marginal utility is based on the following assumptions.

- The consumer is rational
- The consumer consumes identical or homogenous product. The commodity to be consumed should have similar quality, color, design, etc.
- There is no time gap in consumption of the good
- ·The consumer taste/preferences remain unchanged

3.3.1.4 Equilibrium of a consumer

The objective of a rational consumer is to maximize total utility

Given his limited income and the price level of goods and services, What combination of goods and services should he consume so as to get the maximum total utility?

a) the case of one commodity

The equilibrium condition of a consumer that consumes a single good X occurs when the marginal utility of X is equal to its market price.

Proof

Given the utility function

$$U = f(X)$$

If the consumer buys commodity X, then his expenditure will be $Q_X P_X$. The consumer maximizes the difference between his utility and expenditure.

$$Max(U - Q_X P_X)$$

The necessary condition for maximization is equating the derivative of a function to zero. Thus,

$$\frac{dU}{dQ_X} - \frac{d(Q_X P_X)}{dQ_X} = 0$$

$$\frac{dU}{dQ_X} - P_X = 0 \Rightarrow MU_X = P_X$$

b) the case of two or more commodities

For the case of two or more goods, the consumer ‘s equilibrium is achieved when the marginal

utility per money spent is equal for each good purchased and his money income available for the purchase of the goods is exhausted. That is,

$$\frac{MU_X}{P_X} = \frac{MU_Y}{P_Y} = \cdots = \frac{MU_N}{P_N} \quad \text{and} \quad P_X Q_X + P_Y Q_Y + \cdots + P_N Q_N = M$$

Limitation of the cardinal approach

1. The assumption of cardinal utility is doubtful because utility may

not be quantified. Utility cannot be measured absolutely (objectively).

2. The assumption of constant MU of money is unrealistic because as

income increases, the marginal utility of money changes.

3.3.2 The ordinal utility theory

The consumers can rank commodities in the order of their preferences as 1st, 2nd, 3rd and so on.

3.3.2.1 Assumptions of ordinal utility theory

- **Consumers are rational** - they maximize their satisfaction or utility given their income and market prices.
- **Utility is ordinal** - utility is not absolutely (cardinally) measurable. Consumers are required only to order or rank their preference for various bundles of commodities.
- **Diminishing marginal rate of substitution:** The marginal rate of substitution is the rate at which a consumer is willing to substitute one commodity for another commodity so that his total satisfaction remains the same. The rate at which one good can be substituted for another in consumer‘s basket of goods diminishes as the consumer consumes more and more of the good.
- The **total utility** of a consumer is measured by the amount (quantities) of all items he/she consumes from his/her **consumption basket.**
- **Consumer’s preferences are consistent.** For example, if there are three goods in a given consumer ‘s basket, say, X, Y, Z and if he prefers X to Y and Y to Z, then the consumer is expected to prefer X to Z. This property is known as axioms of ***transitivity*****.**

The ordinal utility approach is explained with the help of indifference curves. Therefore, the

ordinal utility theory is also known as the *indifference curve approach.*

3.3.2.2 Indifference set, curve and map

Indifference set/ schedule is a combination of goods for which the consumer is indifferent. It shows the various combinations of goods from which the consumer derives the same level of satisfaction.

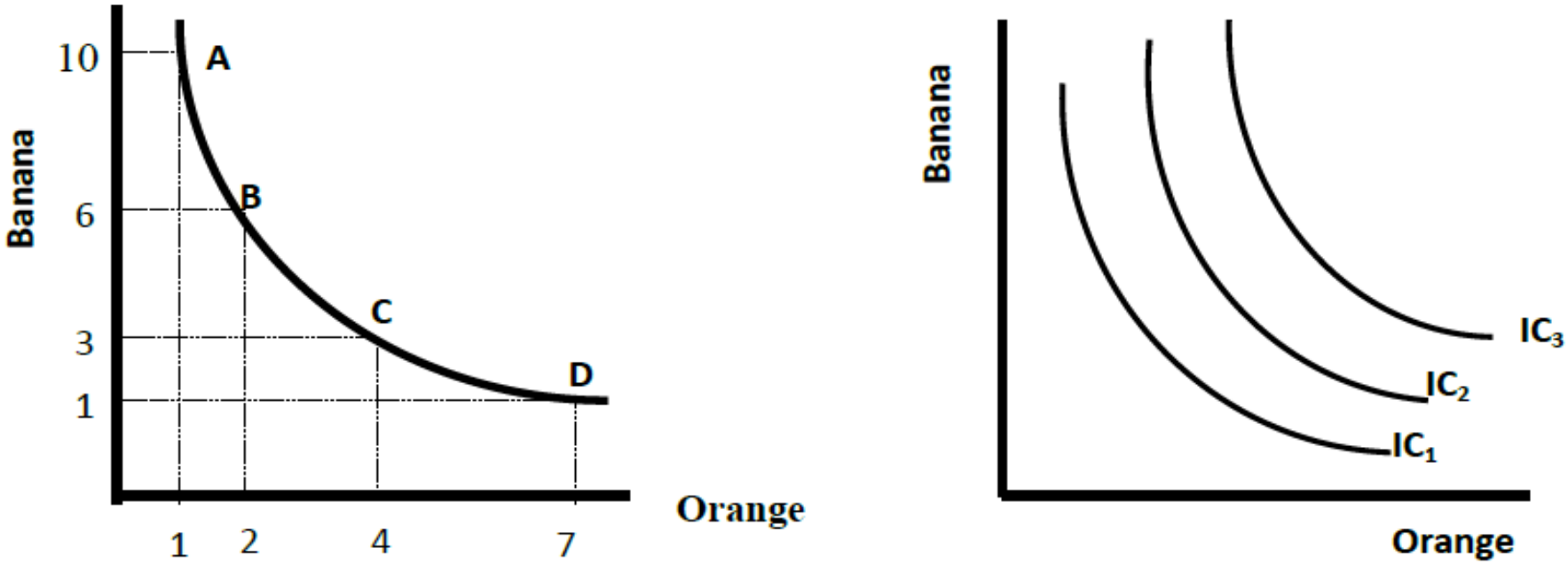
Consider a consumer who consumes two goods X and Y (table 3.3).

Table 3.3: Indifference schedule

Bundle (Combination)	A	B	C	D
Orange	1	2	4	7
Banana	10	6	3	1

In table 3.3 above, each combination of good X and Y gives the consumer equal level of total utility. Thus, the individual is indifferent whether he consumes combination A, B, C or D.

Indifference curve: An indifference curve is curve shows different combinations of two goods which yield the same utility (level of satisfaction) to the consumer. A set of indifference curves is called **indifference map.**



i) Indifference curve ii) Indifference map

Figure 3.3: Indifference curve and indifference map

3.3.2.3 Properties of indifference curves

1. **Indifference curves have negative slope (downward sloping to the right).** Indifference curves are negatively sloped because the consumption level of one commodity can be increased only by reducing the consumption level of the other commodity.
2. **Indifference curves are convex to the origin.** The convexity of indifference curves is the reflection of the diminishing marginal rate of substitution.
3. **A higher indifference curve is always preferred to a lower one.** The further away from the origin an indifferent curve lies, the higher the level of utility it denotes.
4. **Indifference curves never cross each other (cannot intersect).** The assumptions of consistency and transitivity will rule out the intersection of indifference curves.

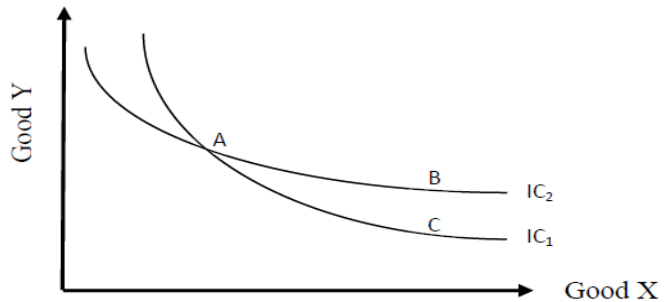


Figure 3.4: Intersection of indifference curves

3.3.2.4 Marginal rate of substitution (MRS)

Marginal rate of substitution is a rate at which consumers are willing to substitute one commodity for another in such a way that the consumer remains on the same indifference curve.

Marginal rate of substitution of X for Y is defined as the number of units of commodity Y that must be given up in exchange for an extra unit of commodity X so that the consumer maintains the same level of satisfaction.

$$MRS_{X,Y} = \frac{Number\ of\ units\ of\ Y\ given\ up}{Number\ of\ units\ of\ X\ gained} = -\frac{\Delta Y}{\Delta X}$$

To understand the concept, consider the following indifference curve.

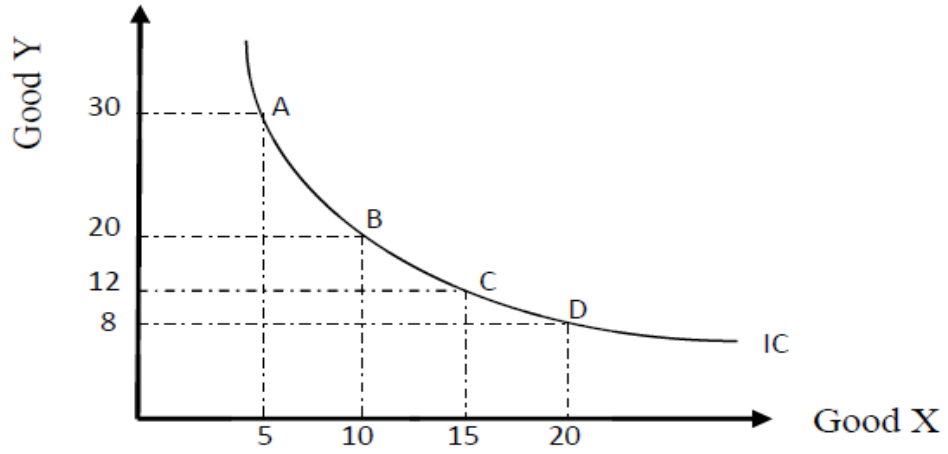


Figure 3.5: Indifference curve for two products X and Y

From the above graph, $MRS_{X,Y}$ associated with the movement from point A to B, point B to C and point C to D is 2.0,1.6, and 0.8 respectively

It is also possible to derive MRS using the concept of marginal utility. $MRS_{X,Y}$ is related to MU_X and MU_Y as follows.

$$MRS_{X,Y} = \frac{MU_X}{MU_Y}$$

Proof: Suppose the utility function for two commodities X and Y is defined as:

$$U = f(X,Y)$$

Since utility is constant along an indifference curve, the total differential of the utility function will be zero.

$$dU = \frac{\partial U}{\partial X}dX + \frac{\partial U}{\partial Y}dY = 0$$

$$MU_X dX + MU_Y dY = 0$$

$$\frac{MU_X}{MU_Y} = -\frac{dY}{dX} = MRS_{X,Y} \qquad \text{Similarly, } \frac{MU_Y}{MU_X} = -\frac{dX}{dY} = MRS_{Y,X}$$

Example: Suppose a consumer's utility function is given by $U(X,Y) = X^4Y^2$. Find $MRS_{X,Y}$

Solution: $MRS_{X,Y} = \frac{MU_X}{MU_Y}$

$$MU_X = \frac{\partial U}{\partial X} = 4X^3Y^2 \quad and\ MU_Y = \frac{\partial U}{\partial Y} = 2X^4Y \quad \text{Hence, } MRS_{X,Y} = \frac{MU_X}{MU_Y} = \frac{4X^3Y^2}{2X^4Y} = \frac{2Y}{X}$$

3.3.2.5 The budget line or the price line

In reality, the consumer is constrained by his/her income and prices of the two commodities. This constraint is often presented with the help of the budget line.

The ***budget line*** is a set of the commodity bundles that can be purchased if the entire income is spent. It is a graph which shows the various combinations of two goods that a consumer can purchase given his/her limited income and the prices of the two goods.

In order to draw a budget line facing a consumer, we consider the following assumptions.

- There are only two goods bought in quantities, say, X and Y.
- Each consumer is confronted with market determined prices, PX and PY.
- The consumer has a known and fixed money income (M).

Assuming that the consumer spends all his/her income on the two goods (X and Y), we can express the budget constraint as:

$$M = P_X X + P_Y Y$$

By rearranging the above equation, we can derive the following general equation of a budget line.

$$Y = \frac{M}{P_Y} - \frac{P_X}{P_Y} X$$

Graphically,

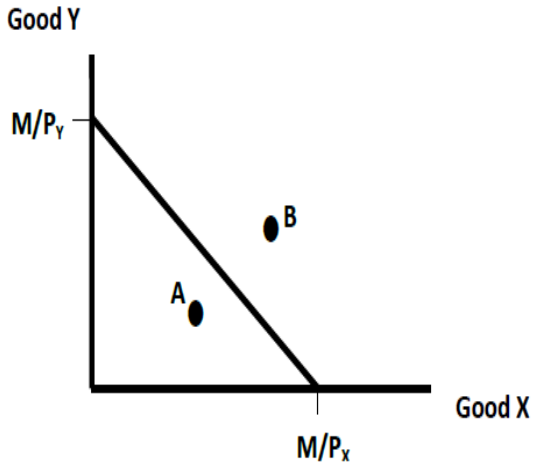


Figure 3.6: The budget line

Note that:

- The slope of the budget line is given is by-$\frac{Px}{Py}$ (the ratio of the prices of the two goods).
- Any combination of the two goods within the budget line (such as point A) or along the budget line is attainable.
- Any combination of the two goods outside the budget line (such as point B) is unattainable (unaffordable).

Example: A consumer has \$100 to spend on two goods X and Y with prices \$3 and \$5 respectively. Derive the equation of the budget line and sketch the graph.

Solution: The equation of the budget line can be derived as follows.

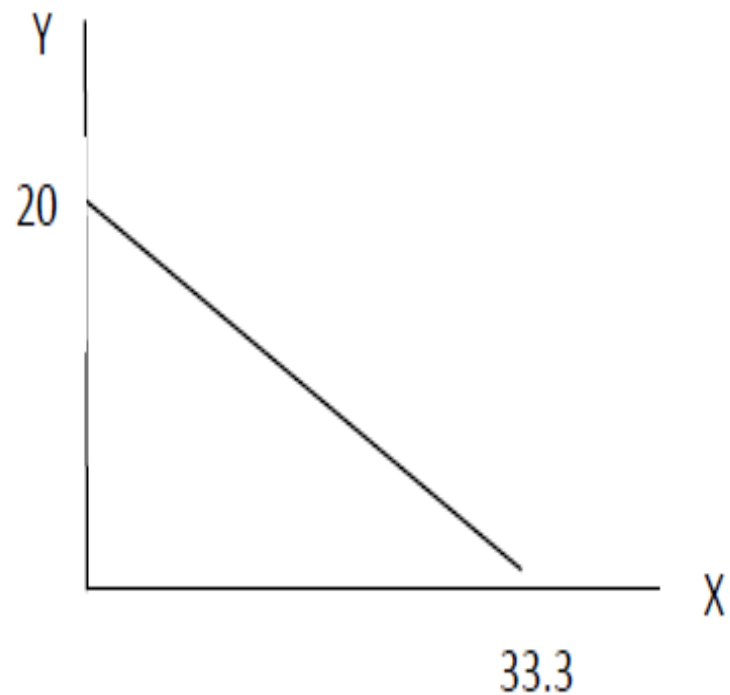
$$P_X X + P_Y Y = M$$

$$3X + 5Y = 100$$

$$5Y = 100 - 3X$$

$$Y = \frac{100}{5} - \frac{3}{5}X$$

$$Y = 20 - \frac{3}{5}X$$



The changes in prices or income will affect the budget line.

Change in income

- Increase in income causes an upward/outward shift
- decreases in income causes a downward/inward shift
- It is important to note that the slope of the budget line (the ratio of the two prices) does not change when income rises or falls.

Change in price : An equal increase in the prices of the two goods shifts the budget line inward. An equal decrease in the prices of the two goods, one the other hand, shifts the budget line out ward.

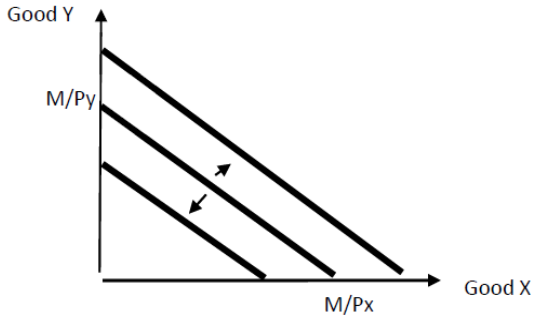


Figure 3.7: *Effects of increase (right) and decrease (left) in income on the budget line*

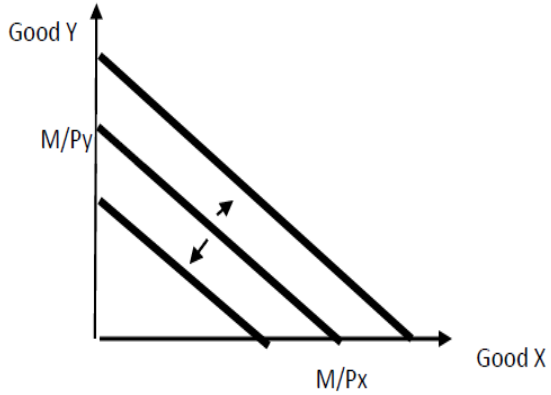


Figure 3.8: *Effect of proportionate increase (inward) and decrease (out ward) in the prices of both goods*

An increase or decrease in the price of one of the two goods, keeping the price of the other good and income constant, changes the slope of the budget line by affecting only the intercept of the commodity that records the change in the price.

For instance, if the price of good X decreases while both the price of good Y and consumer‘s income remain unchanged, the horizontal intercept moves outward and makes the budget line flatter. The reverse is true if the price of good X increases. On the other hand, if the price of good Y decreases while both the price of good X and consumer‘s income remain unchanged, the vertical intercept moves upward and makes the budget line steeper. The reverse is true for an increase in the price of good Y.

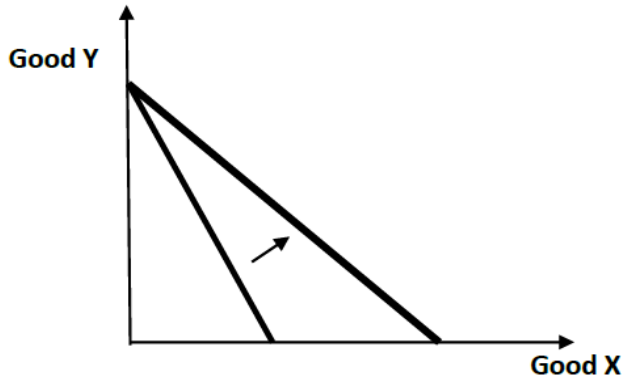


Figure 3.9: Effect of decrease in the price of only good X on the budget line

3.3.2.6 Equilibrium of the consumer

The equilibrium of consumer occurs at the point where the indifference curve is tangent to the budget line so that the slope of the indifference curve (*MRS XY*) is equal to the slope of the budget line (P_x /P_y). In figure 3.10, the equilibrium of the consumer is at point E 'where the budget line is tangent to the highest attainable indifference curve (IC2).

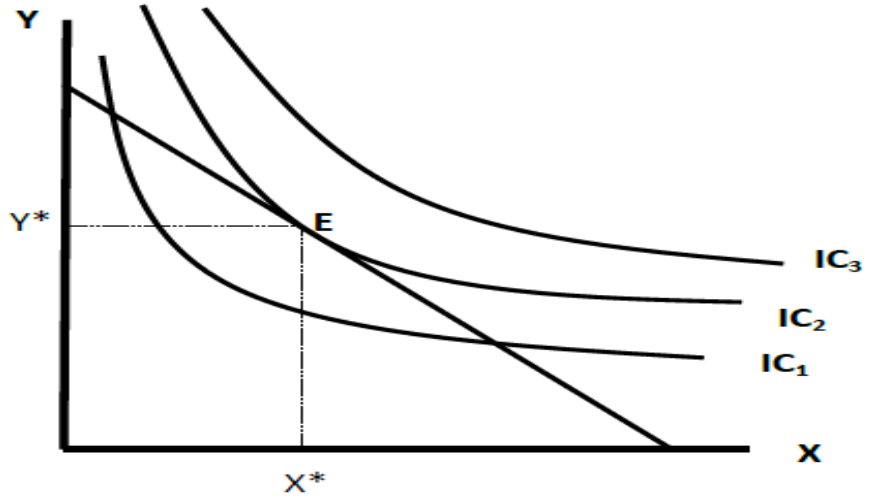


Figure 3.10: Consumer equilibrium under indifference curve approach

Mathematically, consumer optimum (equilibrium) is attained at the point where:

Slope of indifference curve = Slope of the budget line

$$MRS_{XY} = \frac{P_X}{P_Y}$$

$$\Rightarrow \frac{MU_X}{MU_Y} = \frac{P_X}{P_Y}$$

Example: A consumer consuming two commodities X and Y has the utility function $U(X,Y) = XY + 2X$. The prices of the two commodities are 4 birr and 2 birr respectively. The consumer has a total income of 60 birr to be spent on the two goods.

a) Find the utility maximizing quantities of good X and Y.

b) Find the $MRS_{X,Y}$ at equilibrium.

Solution

a) The budget constraint of the consumer is given by:

$P_X.X + P_Y.Y = M$

$4X+2Y= 60$ ..... (i)

Moreover, at equilibrium

$$\frac{MU_X}{MU_Y} = \frac{P_X}{P_Y}$$

$$\frac{Y+2}{X} = \frac{4}{2}$$

$$\frac{Y+2}{X} = 2$$

$$Y = 2X - 2 \quad \text{..... (ii)}$$

Substituting equation (ii) into (i), we obtain $Y = 14 \quad and \quad X = 8.$

b) $MRS_{X,Y} = \frac{MU_X}{MU_Y} = \frac{Y+2}{X} = \frac{14+2}{8} = 2$

(At the equilibrium, MRS can also be calculated as the ratio of the prices of the two goods)